

Faculty of Electrical Engineering

WARSAW UNIVERSITY OF TECHNOLOGY

Control of Energy Conversion systems

PROJECT

TITLE

H4(BIPOLAR, UNIPOLAR) AND H5

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1. Introduction

The primary objective of this project is to analyze, simulate, and compare the performance of full-bridge (H4) and H5 inverter topologies under different Pulse Width Modulation (PWM) strategies. Specifically, the project focuses on evaluating how Bipolar PWM, Unipolar PWM, and the specialized H5 topology affect the generation of Common-Mode Voltage (CMV). Because transformerless photovoltaic (PV) and energy conversion systems suffer from parasitic capacitance between the PV panel and the ground, minimizing CMV is crucial to eliminating dangerous ground leakage currents.

2. Literature review

Transformerless inverters have gained immense popularity in energy conversion due to their high efficiency, lower cost, and compact size. However, the lack of galvanic isolation introduces a direct path for common-mode leakage currents through the system's parasitic capacitances.

Literature indicates that standard full-bridge (H4) inverters operating with Bipolar PWM keep the common-mode voltage constant, which theoretically eliminates leakage current but introduces high output current ripple and lower efficiency. Conversely, Unipolar PWM improves output voltage quality and reduces current ripple, but it introduces high-frequency variations in the CMV, leading to severe leakage currents. To solve this dilemma, topologies like the H5 decouple the DC side from the AC side during freewheeling states, effectively clamping the common-mode voltage and drastically minimizing leakage currents without sacrificing output filter efficiency.

3. Methodology

To investigate the interaction between modulation strategies and inverter topologies, plecs software modeling was used to replicate standard full-bridge configurations and the specialized H5 configuration.

3.1 Mathematical Formulation

The common-mode voltage Vcm of a single-phase H-bridge inverter is defined mathematically as the arithmetic average of the two phase-leg output potentials (Node A, denoted as Va, and Node B, denoted as Vb measured relative to a shared DC reference point:

$$Vcm = (Va + Vb)/2$$

The magnitude and frequency components of the ground leakage current I_leakage flowing through the system's parasitic capacitance Cp are directly governed by the time-domain rate of change of the voltage over capacitor:

$$I_{leakage} = Cp * dVcap/dt$$

3.2 Simulation Configuration

The electrical models were simulated under uniform parameters using a standard DC bus link voltage of Vdc = 400V, L = 2mH, R_load = 5Ohms, C_load = 50mF. C_parasitic = 100pF, R_parasitic = 10kOhms. In the PLECS schematic an extra Load switch can be observed but the results shown and discussed in this report do not apply to that switching stage. That switching stage worked as a tool for me to gain the understanding of switchings. The

performance of three primary configurations were documented:
H4 Bridge with Bipolar modulation
H4 Bridge with Unipolar modulation
H5 Decoupled Inverter Topology

4. Results and Discussion

4.1 Defining and Measuring Common Mode Voltage

As defined by the node voltage calculations, the behavior of Vcm shifts dramatically based on the chosen topology and switching commands:

Bipolar Modulation (H4): As illustrated by the simulation waveforms, the control signals force the diagonal switches to act in strict opposition. When Node A is tied to the positive DC rail Va = 400V, Node B is grounded Vb = 0V, and vice versa. Applying the formula gives a constant value:

$$Vcm = (400 + 0)/2 = 200\ V$$

1st graph - Node A
2nd graph - Node B
3rd graph - Common mode voltage - applies to figures in this subchapter

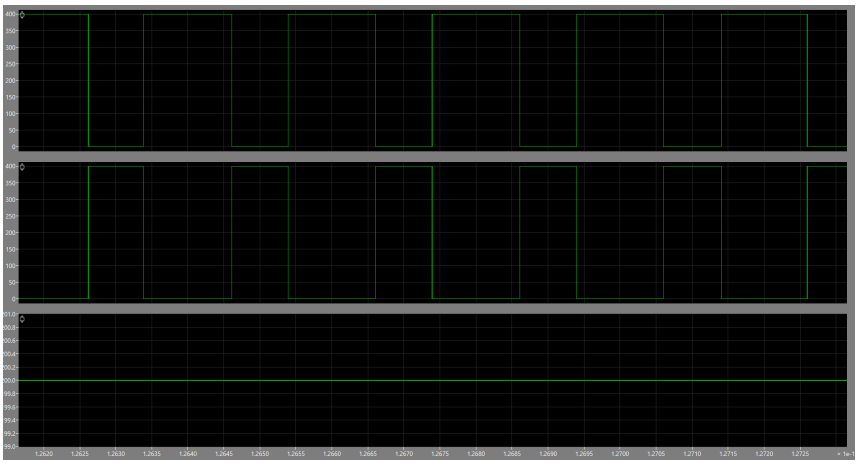


Figure 1 - Bipolar Node A, Node b, common mode voltages

Unipolar Modulation (H4): The phase legs are modulated independently. During active energy transmission intervals, one node is at 400V while the other is at 0V Vcm = 200V. However, during zero-voltage freewheeling intervals, both output nodes are simultaneously clamped to the positive rail Va = Vb = 400V, resulting in Vcm = 400V or the negative rail Va = Vb = 0V, resulting in Vcm = 0V. This induces severe switching-frequency oscillations.

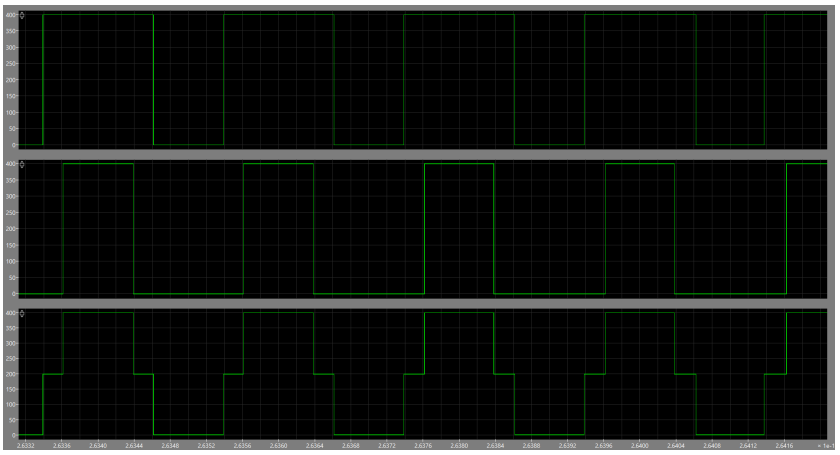


Figure 2 - unipolar Node A, Node b, common mode voltages

H5 Topology: The H5 circuit features a fifth active switch on the positive DC bus rail. During active intervals, it mirrors standard full-bridge operation. During freewheeling states, the H5 switch turns off to completely isolate the DC source from the AC bridge. The lower switches allow the inductor current to circulate safely. Because the DC link is disconnected, the phase node potentials do not collapse to 0V or spike to Vdc relative to the system ground, stabilizing the CMV profiles.

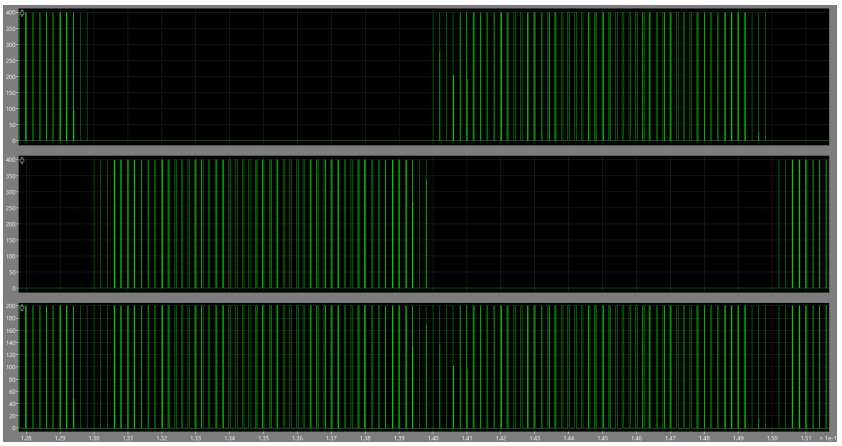


Figure 3 - H5 Node A, Node b, common mode voltages - periodicity

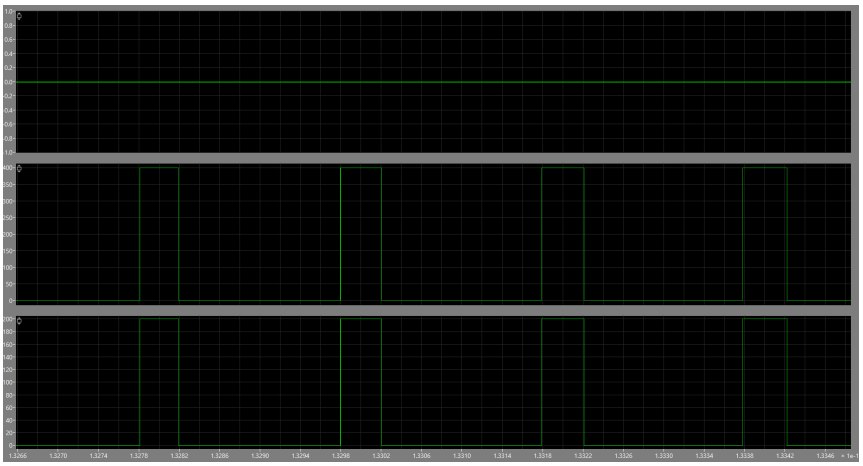


Figure 4 - H5 Node A, Node b, common mode voltages - detailed view

4.2 Switching Characteristics of Common Mode Voltage

The dynamic properties of the measured CMV across the three systems are summarized below:

Modulation	**Amplitude of Vcm**	**Rate of Change (dVcm/dt)**	**Harmonic Spectra Characteristics**
Bipolar (H4)	Constant at 200 V	Zero 0	Pure DC; entirely free of high-frequency peaks
Unipolar (H4)	Full rail-to-rail swing 0V - 400V	Extremely High	Harmonic spikes at higher frequencies
H5 Topology	Square wave between 0V and 200V	medium	Messiest spectrum figure

Table 1

The frequency domain spectrum graphs validate these findings. The Bipolar PWM spectrum yields a blank high-frequency profile due to its flat DC nature. Unipolar PWM displays a distinct harmonic peak at the carrier frequency, showing a high concentration of unwanted high-frequency energy. The H5 topology effectively dampens these high-frequency peaks, demonstrating the success of its DC-decoupling design.

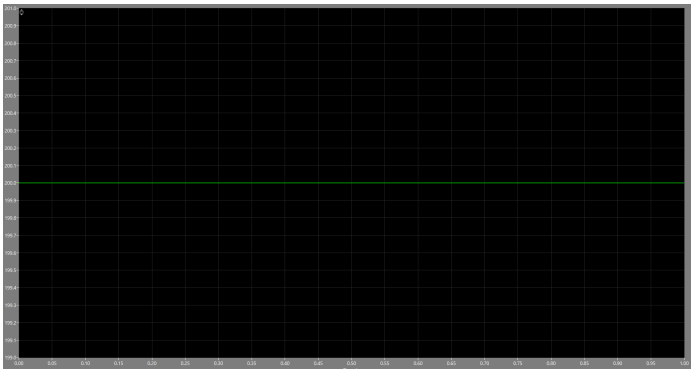


Figure 5 - bipolar common mode voltage



Figure 6 - bipolar frequency content of common mode voltage

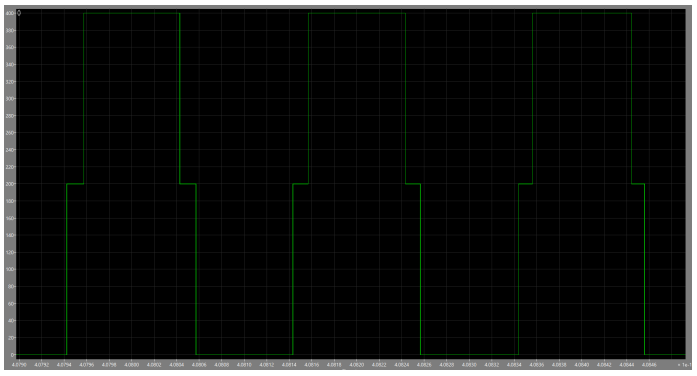


Figure 7 - unipolar common mode voltage

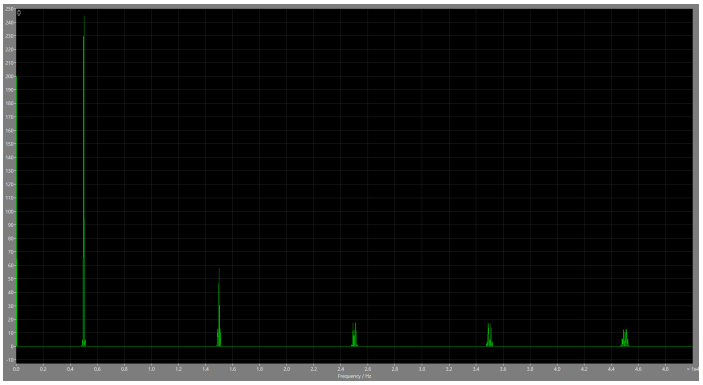


Figure 8 - unipolar frequency content of common mode voltage

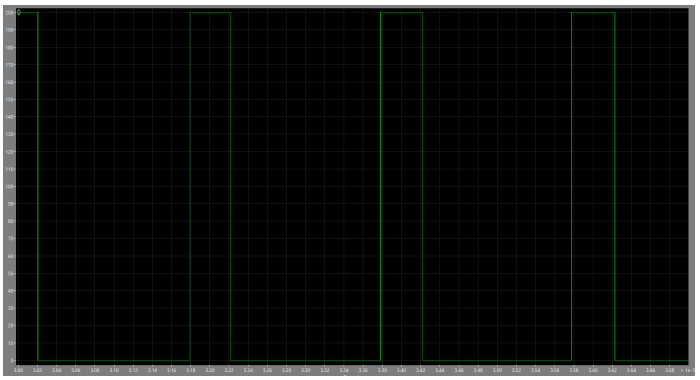


Figure 9 - H5 common mode voltage

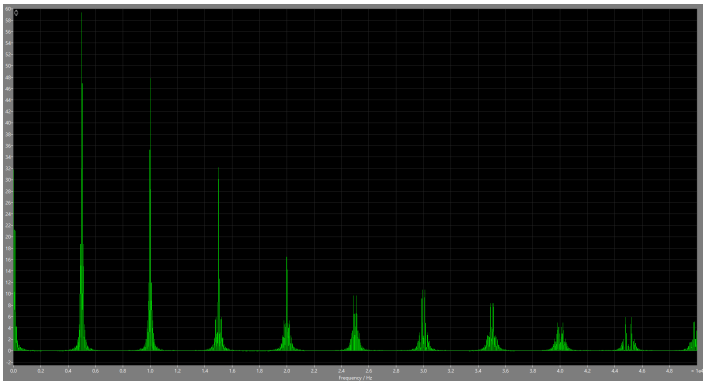


Figure 10 - H5 Frequency spectrum of common mode voltage

4.3 Output Inverter Voltage and Output Inverter Current

The trade-offs of each method are visible when analyzing the output voltage and filter current waveforms:

Bipolar: The output AC voltage switches abruptly between full positive and negative rails +400V to -400V. This severe voltage swing causes a large ripple current in the output filter, requiring larger, heavier inductors to maintain a clean grid current sine wave. Current ripple around 18V.

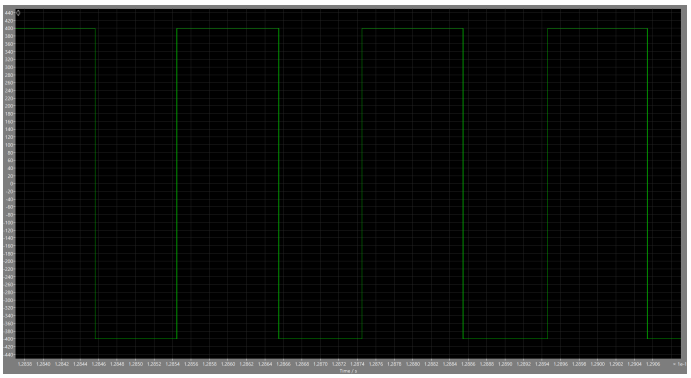


Figure 11- bipolar output voltage

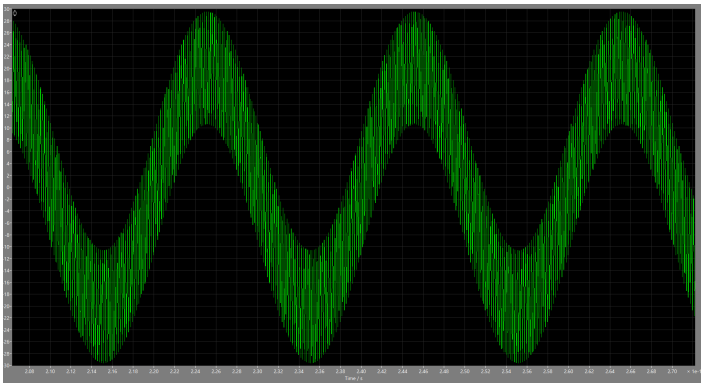


Figure 12 - bipolar output current

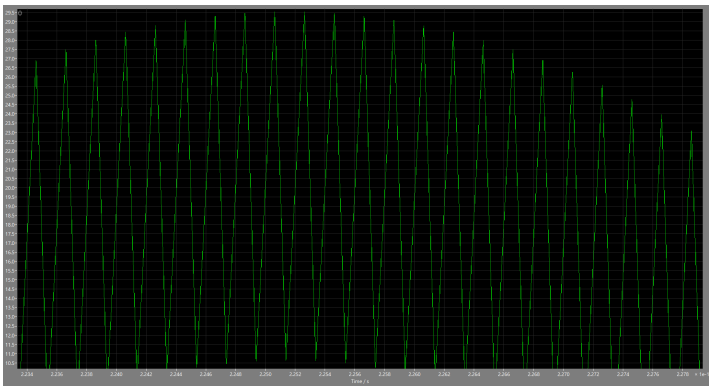


Figure 13 - bipolar output current ripple

Unipolar: The output voltage alternates between +400V and 0V, or -400V and 0V. This three-level modulation effectively doubles the apparent ripple frequency seen by the filter. Current ripple around 3.5V, quite a decrease.

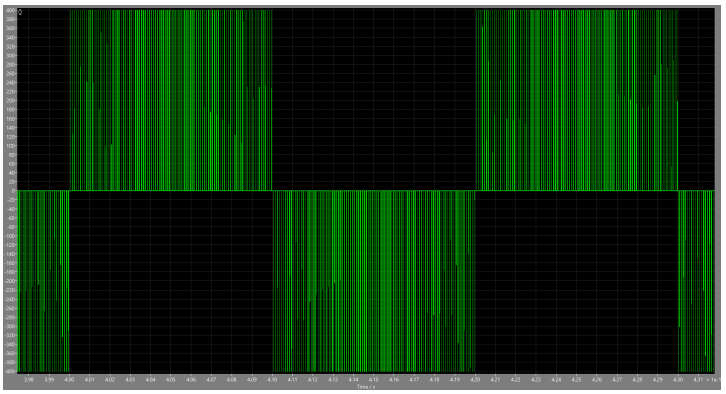


Figure 14 - unipolar output voltage

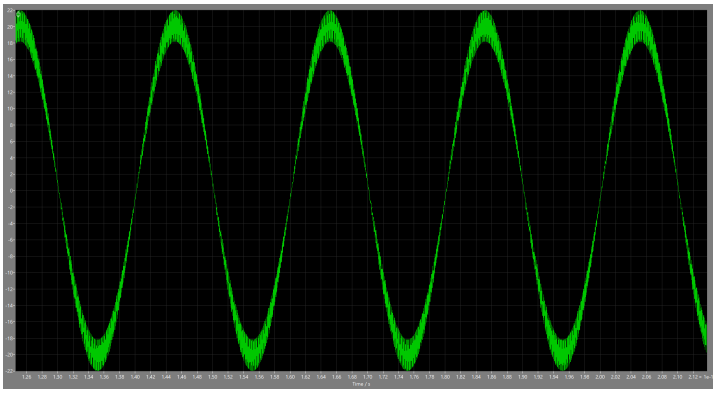


Figure 15 - unipolar output current

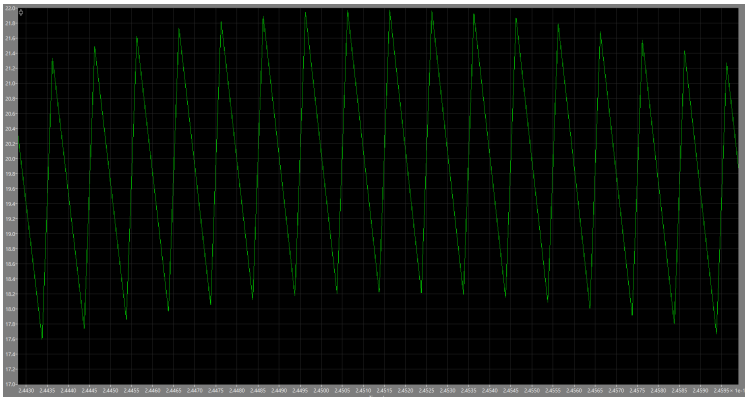


Figure 16 - unipolar output current ripple

H5: The H5 configuration maintains the three-level output voltage profile +Vdc, 0, -Vdc characteristic of unipolar systems. Consequently, its output current current ripple remains minimal, preserving high filter efficiency without necessitating oversized magnetic components. Current ripple around 8V, still a huge difference from bipolar modulation case.

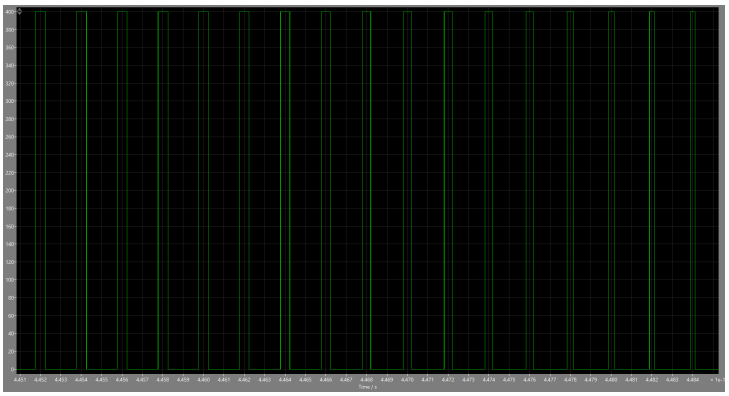


Figure 17 - H5 output voltage change

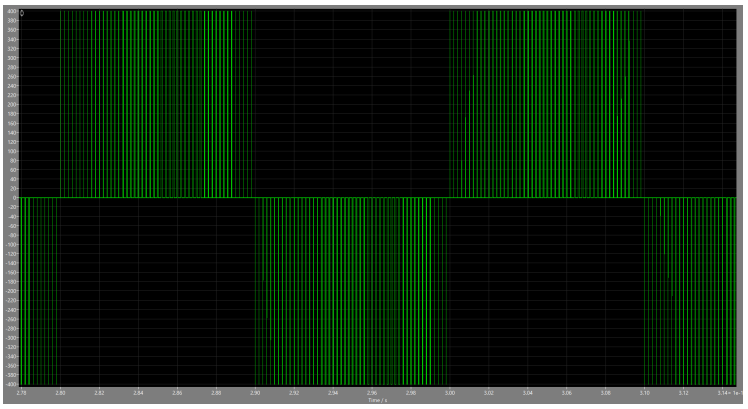


Figure 18 - H5 output voltage range

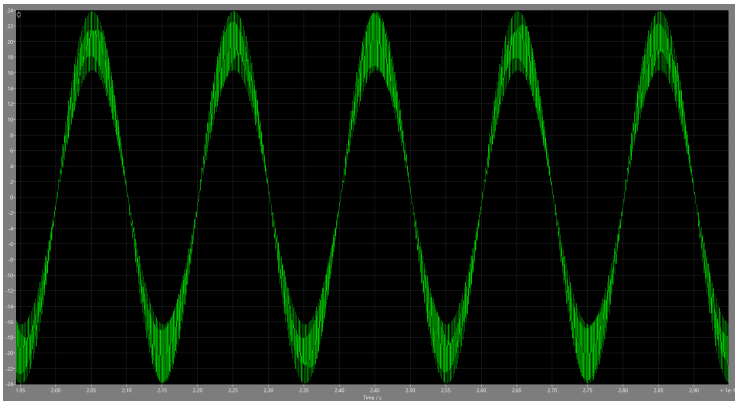


Figure 19 - H5 output current

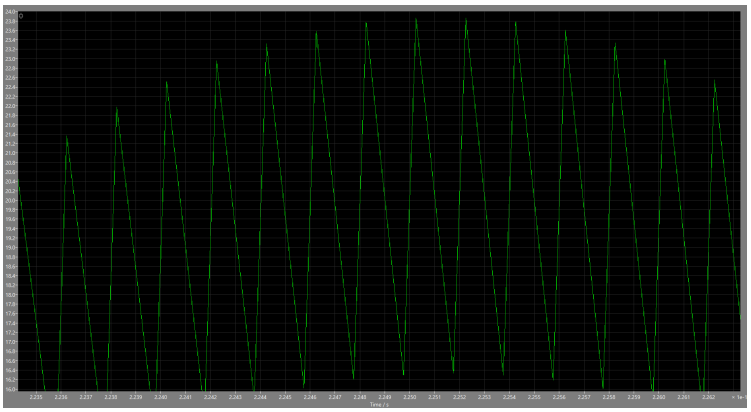


Figure 20 - H5 output current ripple

4.4 Estimation and Discussion of Resulting Leakage Current

Applying the relationship I_leakage = Cp * dVcm/dt, the simulation results map the ground leakage currents directly to the behavior of the voltage:

Bipolar: Since the voltage is flat is approx 0, the ground leakage current remains near zero, showing only negligible, transient spikes during dead-time transitions.

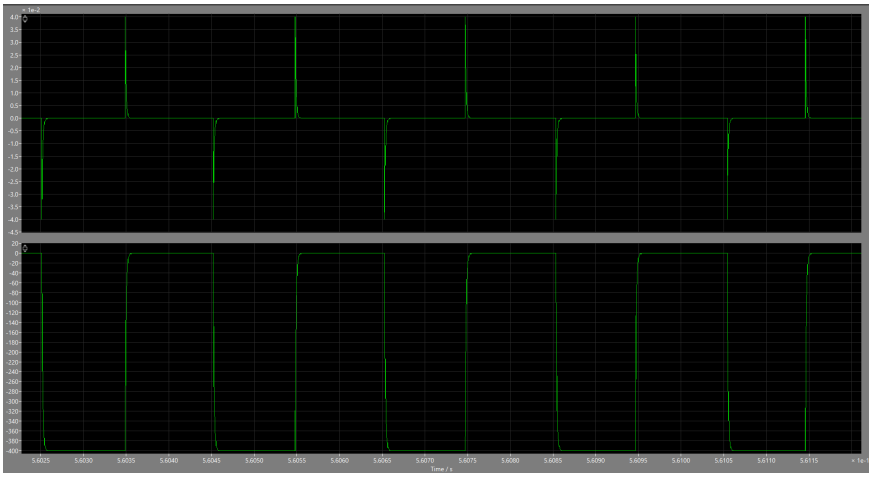


Figure 21 - leakage current and capacitance voltage

Unipolar: The current spikes are almost the same as in previous case, except the spikes to the positive side right after negative spikes. And the frequencies are the it is not that this current ripples has more frequency, meaning they are more. But, indeed it si more in this case due to the spikes i mentioned before.

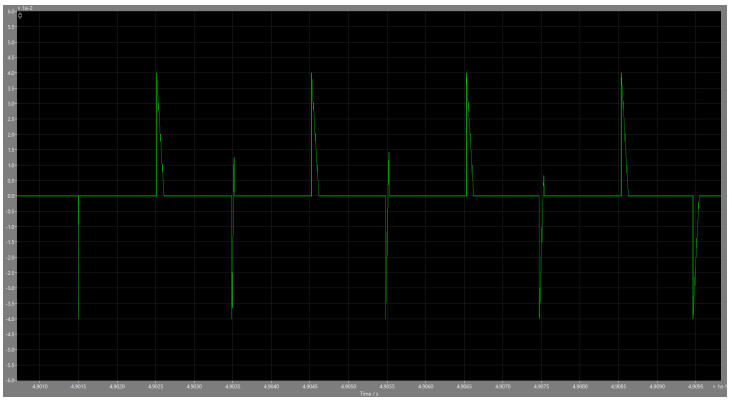


Figure 22 - Leakage current

H5: By opening the fifth switch during freewheeling states, the H5 topology breaks the path of the leakage current through the ground loop. The simulation results demonstrate that the leakage current spikes are well-insulated and suppressed, keeping leakage current to only half of the time(period).

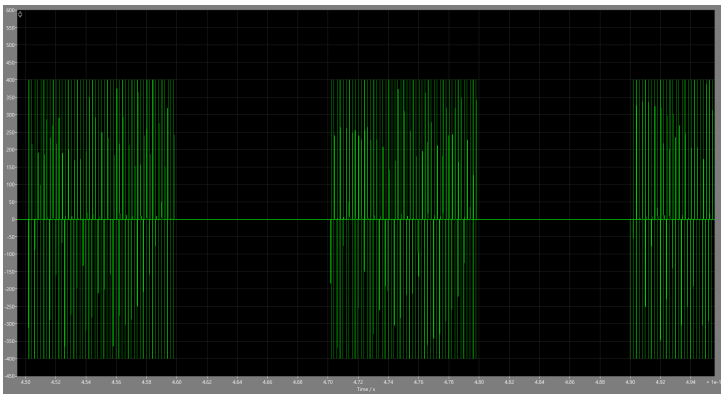


Figure 24 - Leakage current periodicity

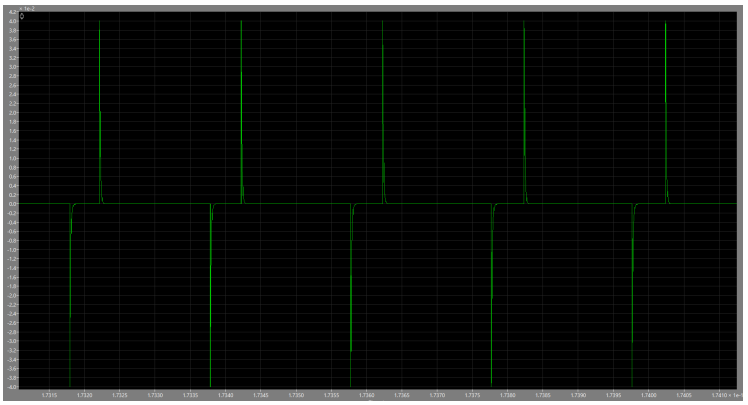


Figure 25 - Leakage current

4.5 Topology and PWM Strategy Influences on EMI and System Performance

The choice of topology directly dictates the system's electromagnetic interference (EMI) profile.

In a standard H4 inverter, Bipolar PWM delivers excellent CMV behavior but suffers from high switching stress and severe core losses within the filter inductors. Conversely, Unipolar PWM features low inductor stress and high efficiency but produces severe high-frequency CMV variations. This creates an analytical trade-off between filter size, conversion losses, and EMI emission.

The H5 topology serves as a highly effective solution. By introducing an active decoupling switch, it preserves the high-efficiency, low-ripple characteristics of three-level unipolar switching on the AC output side, while forcing the CMV to behave smoothly. This results in a minimal EMI footprint, eliminates the need for bulky common-mode filters, and ensures high overall system performance.

5. Conclusions

5.1 Achievements and Goals

All primary goals established for this project were successfully realized. The full-bridge H4 (Bipolar and Unipolar) and H5 power electronics circuits were accurately modeled and analyzed using simulation software. The waveforms confirmed the theoretical performance characteristics: Bipolar

modulation minimizes leakage current at the cost of high output ripple, Unipolar modulation minimizes output ripple at the cost of high leakage current, and the H5 topology combines the advantages of both systems.

5.2 Challenges and Troubleshooting

The primary technical challenge involved properly modeling the modulation for H5 which isnt directly stated in any scientific articles. Probably considered obvious or to be determined by the readers himself. So, in-depth understanding was required to analyze the modulated s signals which control the semiconductors and from them derive the modulation for S5, fifth switch, and others. The next challenge was to collect the leakage current which is not really part of the circuit in simulation but exists in real life circuits of this type. Which was looked up from the [1] report by Amrutha Daibagna.

References

1.Design of H5 Transformerless Inverter for Photovoltaic System
https://iopscience.iop.org/article/10.1088/1757-899X/1295/1/012016/pdf
2. Laboratory materials 1,2